```
# Zajęcie 1 – Nauka o danych
# Wariant 4: United States Healthcare Spending by Race and Ethnicity 2002–2016

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import zipfile
import io

print("Biblioteki zostały zaimportowane poprawnie.")
print("Wersja pandas:", pd.__version__)
```

```
Biblioteki zostały zaimportowane poprawnie.
Wersja pandas: 2.2.3
```

```
from google.colab import files

uploaded = files.upload()
```

Wybierz pliki IHME_USA…16_DATA.zip
IHME_USA_HEALTH_SPENDING_RACE_ETHNICITY_2002_2016_DATA.zip(application/x-zip-compressed) - 6437178 bytes, last modified: 17.10.2026 - 100% done

```
# Odczyt pliku CSV bezpośrednio z archiwum ZIP

zip_filename = next(iter(uploaded))

with zipfile.ZipFile(io.BytesIO(uploaded[zip_filename])) as archive:
    csv_files = [
        name for name in archive.namelist()
        if name.lower().endswith(".csv")
    ]

    print("Pliki CSV w archiwum:")
    for name in csv_files:
        print(name)

    data_filename = csv_files[0]

    with archive.open(data_filename) as file:
        df = pd.read_csv(file, low_memory=False)

print("\nDane zostały wczytane.")
print("Liczba wierszy i kolumn:", df.shape)
```

```
Pliki CSV w archiwum:
IHME_USA_HEALTH_SPENDING_RACE_ETHNICITY_2002_2016_DATA_Y2021M08D17.CSV

Dane zostały wczytane.
Liczba wierszy i kolumn: (598500, 13)
```

```
display(df.head(10))
```

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care	spending_unit	v
0	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Ambulatory Care	Spending in 2016 USD (millions)	104
1	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Dental Care	Spending in 2016 USD (millions)	3
2	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Emergency Department Care	Spending in 2016 USD (millions)	24
3	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Inpatient Care	Spending in 2016 USD (millions)	24
4	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Nursing Facility Care	Spending in 2016 USD (millions)	2
5	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Prescribed Pharmaceuticals	Spending in 2016 USD (millions)	20
6	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Total	Spending in 2016 USD (millions)	180
7	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	cvd	Cardiovascular diseases	Ambulatory Care	Spending in 2016 USD (millions)	1
8	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	cvd	Cardiovascular diseases	Emergency Department Care	Spending in 2016 USD (millions)	0
9	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	cvd	Cardiovascular diseases	Inpatient Care	Spending in 2016 USD (millions)	1

```
# Wyświetlenie ostatnich 10 wierszy
display(df.tail(10))

# Informacje o ramce danych
df.info()

# Liczba wierszy i kolumn
print("Liczba wierszy:", df.shape[0])
print("Liczba kolumn:", df.shape[1])

# Nazwy kolumn
print("\nNazwy kolumn:")
print(df.columns.tolist())
```

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care	spending
598490	2016	22	All Ages	1	Male	White (non-Hispanic)	resp_copd	COPD	Inpatient Care	Spend capita i
598491	2016	22	All Ages	1	Male	White (non-Hispanic)	resp_copd	COPD	Nursing Facility Care	Spend capita i
598492	2016	22	All Ages	1	Male	White (non-Hispanic)	resp_copd	COPD	Prescribed Pharmaceuticals	Spend capita i
598493	2016	22	All Ages	1	Male	White (non-Hispanic)	resp_copd	COPD	Total	Spend capita i
598494	2016	22	All Ages	1	Male	White (non-Hispanic)	rf_hypertension	Hypertension	Ambulatory Care	Spend capita i
598495	2016	22	All Ages	1	Male	White (non-Hispanic)	rf_hypertension	Hypertension	Emergency Department Care	Spend capita i
598496	2016	22	All Ages	1	Male	White (non-Hispanic)	rf_hypertension	Hypertension	Inpatient Care	Spend capita i
598497	2016	22	All Ages	1	Male	White (non-Hispanic)	rf_hypertension	Hypertension	Nursing Facility Care	Spend capita i
598498	2016	22	All Ages	1	Male	White (non-Hispanic)	rf_hypertension	Hypertension	Prescribed Pharmaceuticals	Spend capita i
598499	2016	22	All Ages	1	Male	White (non-Hispanic)	rf_hypertension	Hypertension	Total	Spend capita i

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 598500 entries, 0 to 598499
Data columns (total 13 columns):
 #   Column          Non-Null Count   Dtype  
---  ------          --------------   -----  
 0   year            598500 non-null  int64  
 1   age_group_id    598500 non-null  int64  
 2   age_group_name  598500 non-null  object 
 3   sex_id          598500 non-null  int64  
 4   sex             598500 non-null  object 
 5   race_ethnicity  598500 non-null  object 
 6   acause          598500 non-null  object 
 7   acause_name     598500 non-null  object 
 8   type_care       598500 non-null  object 
 9   spending_unit   598500 non-null  object 
 10  val             598500 non-null  float64
 11  upper           598500 non-null  float64
 12  lower           598500 non-null  float64
dtypes: float64(3), int64(3), object(7)
memory usage: 59.4+ MB
Liczba wierszy: 598500
Liczba kolumn: 13

Nazwy kolumn:
['year', 'age_group_id', 'age_group_name', 'sex_id', 'sex', 'race_ethnicity', 'acause', 'acause_name', 'type_care', 'spendir
```

```
# Statystyki opisowe kolumn liczbowych
display(df.describe())

# Statystyki opisowe wszystkich kolumn,
# również kategoryzowanych
display(df.describe(include="all"))
```

	year	age_group_id	sex_id	val	upper	lower
count	598500.000000	598500.000000	598500.000000	5.985000e+05	5.985000e+05	5.985000e+05
mean	2009.000000	21.955789	2.000000	1.236938e+03	1.390616e+03	1.112925e+03
std	4.320497	32.813249	0.816497	1.818914e+04	1.841754e+04	1.799055e+04
min	2002.000000	5.000000	1.000000	0.000000e+00	0.000000e+00	0.000000e+00
25%	2005.000000	9.000000	1.000000	3.090000e+00	7.000000e+00	0.000000e+00
50%	2009.000000	14.000000	2.000000	2.626000e+01	4.500000e+01	9.480000e+00
75%	2013.000000	19.000000	3.000000	1.990000e+02	2.869350e+02	1.210000e+02
max	2016.000000	160.000000	3.000000	2.440175e+06	2.440175e+06	2.440175e+06

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care
count	598500.000000	598500.000000	598500	598500.000000	598500	598500	598500	598500	598500
unique	NaN	NaN	20	NaN	3	7	8	8	7
top	NaN	NaN	10 to 14	NaN	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Ambulatory Care
freq	NaN	NaN	30870	NaN	199500	85500	88200	88200	97650
mean	2009.000000	21.955789	NaN	2.000000	NaN	NaN	NaN	NaN	NaN
std	4.320497	32.813249	NaN	0.816497	NaN	NaN	NaN	NaN	NaN
min	2002.000000	5.000000	NaN	1.000000	NaN	NaN	NaN	NaN	NaN
25%	2005.000000	9.000000	NaN	1.000000	NaN	NaN	NaN	NaN	NaN
50%	2009.000000	14.000000	NaN	2.000000	NaN	NaN	NaN	NaN	NaN
75%	2013.000000	19.000000	NaN	3.000000	NaN	NaN	NaN	NaN	NaN
max	2016.000000	160.000000	NaN	3.000000	NaN	NaN	NaN	NaN	NaN

```
# Liczba brakujących wartości w każdej kolumnie
print("Brakujące wartości:")
display(df.isnull().sum())

# Łączna liczba brakujących wartości
print("Łącznie brakujących wartości:", df.isnull().sum().sum())

# Utworzenie kopii bez wierszy zawierających braki
df_clean = df.dropna().copy()

print("Rozmiar przed usunięciem braków:", df.shape)
print("Rozmiar po usunięciu braków:", df_clean.shape)
```

```
Brakujące wartości:
```

	0
year	0
age_group_id	0
age_group_name	0
sex_id	0
sex	0
race_ethnicity	0
acause	0
acause_name	0
type_care	0
spending_unit	0
val	0
upper	0
lower	0

dtype: int64

```
Łącznie brakujących wartości: 0
Rozmiar przed usunięciem braków: (598500, 13)
Rozmiar po usunięciu braków: (598500, 13)
```

```
# Ramka danych utworzona ze słownika
sample_dict = {
    "Rok": [2002, 2003, 2004],
    "Wydatki_USD": [1200, 1350, 1420],
    "Grupa": ["A", "B", "A"]
}

df_dict = pd.DataFrame(sample_dict)

print("Ramka danych ze słownika:")
display(df_dict)


# Ramka danych utworzona z listy list
sample_list = [
    [2002, 1200, "A"],
    [2003, 1350, "B"],
    [2004, 1420, "A"]
]

df_list = pd.DataFrame(
    sample_list,
    columns=["Rok", "Wydatki_USD", "Grupa"]
)

print("Ramka danych z listy list:")
display(df_list)


# Transponowanie ramki danych
print("Ramka danych po transponowaniu:")
display(df_dict.T)
```

Ramka danych ze słownika:

	Rok	Wydatki_USD	Grupa
0	2002	1200	A
1	2003	1350	B
2	2004	1420	A

Ramka danych z listy list:

	Rok	Wydatki_USD	Grupa
0	2002	1200	A
1	2003	1350	B
2	2004	1420	A

Ramka danych po transponowaniu:

	0	1	2
Rok	2002	2003	2004
Wydatki_USD	1200	1350	1420
Grupa	A	B	A

Kolejne kroki: Wygeneruj kod za pomocą zmiennej `df_dict` | New interactive sheet | Wygeneruj kod za pomocą zmiennej `df_list` | New interactiv

```
# Wybór jednej kolumny
display(df["race_ethnicity"].head(10))

# Wybór kilku kolumn
display(df[["year", "race_ethnicity", "type_care", "val"]].head(10))

# Wybór wierszy według indeksów
display(df.iloc[0:5])

# Wybór określonych wierszy i kolumn według indeksów
display(df.iloc[0:5, 0:4])

# Wybór wierszy i kolumn według nazw
display(df.loc[0:4, ["year", "race_ethnicity", "val"]])
```

	race_ethnicity
0	American Indian, Alaska Native (non-Hispanic)
1	American Indian, Alaska Native (non-Hispanic)
2	American Indian, Alaska Native (non-Hispanic)
3	American Indian, Alaska Native (non-Hispanic)
4	American Indian, Alaska Native (non-Hispanic)
5	American Indian, Alaska Native (non-Hispanic)
6	American Indian, Alaska Native (non-Hispanic)
7	American Indian, Alaska Native (non-Hispanic)
8	American Indian, Alaska Native (non-Hispanic)
9	American Indian, Alaska Native (non-Hispanic)

dtype: object

	year	race_ethnicity	type_care	val
0	2002	American Indian, Alaska Native (non-Hispanic)	Ambulatory Care	104.39
1	2002	American Indian, Alaska Native (non-Hispanic)	Dental Care	3.04
2	2002	American Indian, Alaska Native (non-Hispanic)	Emergency Department Care	24.87
3	2002	American Indian, Alaska Native (non-Hispanic)	Inpatient Care	24.94
4	2002	American Indian, Alaska Native (non-Hispanic)	Nursing Facility Care	2.02
5	2002	American Indian, Alaska Native (non-Hispanic)	Prescribed Pharmaceuticals	20.95
6	2002	American Indian, Alaska Native (non-Hispanic)	Total	180.21
7	2002	American Indian, Alaska Native (non-Hispanic)	Ambulatory Care	1.66
8	2002	American Indian, Alaska Native (non-Hispanic)	Emergency Department Care	0.21
9	2002	American Indian, Alaska Native (non-Hispanic)	Inpatient Care	1.59

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care	spending_unit	val	u
0	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Ambulatory Care	Spending in 2016 USD (millions)	104.39	1
1	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Dental Care	Spending in 2016 USD (millions)	3.04	
2	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Emergency Department Care	Spending in 2016 USD (millions)	24.87	
3	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Inpatient Care	Spending in 2016 USD (millions)	24.94	
4	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Nursing Facility Care	Spending in 2016 USD (millions)	2.02	

	year	age_group_id	age_group_name	sex_id
0	2002	5	1 to 4	3
1	2002	5	1 to 4	3
2	2002	5	1 to 4	3
3	2002	5	1 to 4	3
4	2002	5	1 to 4	3

	year	race_ethnicity	val
0	2002	American Indian, Alaska Native (non-Hispanic)	104.39
1	2002	American Indian, Alaska Native (non-Hispanic)	3.04
2	2002	American Indian, Alaska Native (non-Hispanic)	24.87
3	2002	American Indian, Alaska Native (non-Hispanic)	24.94
4	2002	American Indian, Alaska Native (non-Hispanic)	2.02

```
# Rekordy dotyczące roku 2016
df_2016 = df[df["year"] == 2016]
display(df_2016.head(10))

# Sprawdzenie dostępnych wartości w kolumnie sex
print("Dostępne wartości w kolumnie sex:")
print(df["sex"].unique())

# Rekordy dotyczące roku 2016 i płci męskiej
df_2016_male = df[
    (df["year"] == 2016) &
    (df["sex"] == "Male")
]
display(df_2016_male.head(10))

# Wydatki większe od średniej w kolumnie val
df_above_mean = df[df["val"] > df["val"].mean()]
display(df_above_mean.head(10))

print("Liczba rekordów z roku 2016:", len(df_2016))
print("Liczba rekordów z roku 2016 dla mężczyzn:", len(df_2016_male))
print("Liczba rekordów powyżej średniej:", len(df_above_mean))
```

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care	spending_unit
4214	2016	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Ambulatory Care	Spending in 2016 USD (millions)
4215	2016	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Dental Care	Spending in 2016 USD (millions)
4216	2016	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Emergency Department Care	Spending in 2016 USD (millions)
4217	2016	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Inpatient Care	Spending in 2016 USD (millions)
4218	2016	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Nursing Facility Care	Spending in 2016 USD (millions)
4219	2016	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Prescribed Pharmaceuticals	Spending in 2016 USD (millions)
4220	2016	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Total	Spending in 2016 USD (millions)
4221	2016	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	cvd	Cardiovascular diseases	Ambulatory Care	Spending in 2016 USD (millions)
4222	2016	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	cvd	Cardiovascular diseases	Emergency Department Care	Spending in 2016 USD (millions)

```
# Sprawdzenie dostępnych grup rasowych i etnicznych
print(df["race_ethnicity"].unique())

# Wyszukiwanie rekordów zawierających słowo "White"
df_white = df[
    df["race_ethnicity"].astype(str).str.contains(
        "White", case=False, na=False
    )
]
display(df_white.head(10))

# Wyszukiwanie rekordów niezawierających słowa "White"
df_not_white = df[
    ~df["race_ethnicity"].astype(str).str.contains(
        "White", case=False, na=False
    )
]
display(df_not_white.head(10))

print("Liczba rekordów zawierających White:", len(df_white))
print("Liczba rekordów niezawierających White:", len(df_not_white))
```

403219	2016	5	1 to 4	1	Male	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Prescribed Pharmaceuticals	Spending in 2016 USD (millions)
403220	2016	5	1 to 4	1	Male	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Total	Spending in 2016 USD (millions)
403221	2016	5	1 to 4	1	Male	American Indian, Alaska Native (non-Hispanic)	cvd	Cardiovascular diseases	Ambulatory Care	Spending in 2016 USD (millions)
403222	2016	5	1 to 4	1	Male	American Indian, Alaska Native (non-Hispanic)	cvd	Cardiovascular diseases	Emergency Department Care	Spending in 2016 USD (millions)
403223	2016	5	1 to 4	1	Male	American Indian, Alaska Native (non-Hispanic)	cvd	Cardiovascular diseases	Inpatient Care	Spending in 2016 USD (millions)

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care	spending_unit
92	2002	5	1 to 4	3	Both	Black (non-Hispanic)	all	All conditions	Total	Spending in 2016 USD (millions)
129	2002	5	1 to 4	3	Both	Hispanic	all	All conditions	Ambulatory Care	Spending in 2016 USD

```
['American Indian, Alaska Native (non-Hispanic)'
 'Asian, Native Hawaiian, Pacific Islander (non-Hispanic)'
 'Black (non-Hispanic)' 'Hispanic' 'Multiple races (non-Hispanic)' 'Total'
 'White (non-Hispanic)']
```

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care	spending_unit
258	2002	5	1 to 4	3	Both	White (non-Hispanic)	all	All conditions	Ambulatory Care	Spending in 2016 USD (millions)
217	2002	5	1 to 4	3	Both	Total	all	All conditions	Department	Spending in 2016 USD
259	2002	5	1 to 4	3	Both	White (non-Hispanic)	all	All conditions	Dental Care	Spending in 2016 USD (millions)
260	2002	5	1 to 4	3	Both	White (non-Hispanic)	all	All conditions	Emergency Department Care	Spending in 2016 USD (millions)
220	2002	5	1 to 4	3	Both	Total	all	All conditions	Prescribed Pharmaceuticals	Spending in 2016 USD
261	2002	5	1 to 4	3	Both	White (non-Hispanic)	all	All conditions	Inpatient Care	Spending in 2016 USD (millions)
262	2002	5	1 to 4	3	Both	White (non-Hispanic)	all	All conditions	Nursing Facility Care	Spending in 2016 USD (millions)
245	2002	5	1 to 4	3	Both	Total	resp_asthma	Asthma	Total	Spending in 2016 USD
263	2002	5	1 to 4	3	Both	White (non-Hispanic)	all	All conditions	Prescribed Pharmaceuticals	Spending in 2016 USD (millions)
264	2002	5	1 to 4	3	Both	White (non-Hispanic)	all	All conditions	Total	Spending in 2016 USD (millions)

```
Liczba rekordów z roku 2016 dla mężczyzn: 13300
Liczba rekordów powyżej średniej: 56653
```

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care	spending_unit
265	2002	5	1 to 4	3	Both	White (non-Hispanic)	cvd	Cardiovascular diseases	Ambulatory Care	Spending in 2016 USD (millions)
266	2002	5	1 to 4	3	Both	White (non-Hispanic)	cvd	Cardiovascular diseases	Emergency Department Care	Spending in 2016 USD (millions)
267	2002	5	1 to 4	3	Both	White (non-Hispanic)	cvd	Cardiovascular diseases	Inpatient Care	Spending in 2016 USD (millions)

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care	spending_unit	v
0	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Ambulatory Care	Spending in 2016 USD (millions)	104
1	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Dental Care	Spending in 2016 USD (millions)	3
2	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Emergency Department Care	Spending in 2016 USD (millions)	24
3	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Inpatient Care	Spending in 2016 USD (millions)	24
4	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Nursing Facility Care	Spending in 2016 USD (millions)	2
5	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Prescribed Pharmaceuticals	Spending in 2016 USD (millions)	20
6	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Total	Spending in 2016 USD (millions)	180
7	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	cvd	Cardiovascular diseases	Ambulatory Care	Spending in 2016 USD (millions)	1
8	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	cvd	Cardiovascular diseases	Emergency Department Care	Spending in 2016 USD (millions)	0
9	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	cvd	Cardiovascular diseases	Inpatient Care	Spending in 2016 USD (millions)	1

```
Liczba rekordów zawierających White: 85500
Liczba rekordów niezawierających White: 513000
```

```
# Pracujemy na kopii, aby nie zmieniać oryginalnego df
df_modified = df.copy()

# Dodanie nowej kolumny: wartość w tysiącach USD
df_modified["val_thousands"] = df_modified["val"] / 1000

# Usunięcie dodanej kolumny
df_modified = df_modified.drop(columns=["val_thousands"])

# Zmiana nazwy kolumny
df_modified = df_modified.rename(
    columns={"val": "spending_value"}
)

print("Kolumny po modyfikacji:")
print(df_modified.columns.tolist())

display(df_modified.head(5))
```

```
Kolumny po modyfikacji:
['year', 'age_group_id', 'age_group_name', 'sex_id', 'sex', 'race_ethnicity', 'acause', 'acause_name', 'type_care', 'spendir
```

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care	spending_unit	spending_
0	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Ambulatory Care	Spending in 2016 USD (millions)	
1	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Dental Care	Spending in 2016 USD (millions)	
2	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Emergency Department Care	Spending in 2016 USD (millions)	
3	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Inpatient Care	Spending in 2016 USD (millions)	
4	2002	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	all	All conditions	Nursing Facility Care	Spending in 2016 USD (millions)	

```
# Statystyki wydatków
print("Średnia wartość wydatków:", df["val"].mean())
print("Minimalna wartość wydatków:", df["val"].min())
print("Maksymalna wartość wydatków:", df["val"].max())
```

```
Średnia wartość wydatków: 1236.9383054302423
Minimalna wartość wydatków: 0.0
Maksymalna wartość wydatków: 2440174.91
```

```
# Liczba wierszy w ramce danych
print("Liczba rekordów:", len(df))

# Unikatowe lata
print("Lata w zbiorze:")
print(sorted(df["year"].unique()))

# Unikatowe grupy rasowe i etniczne
print("Grupy rasowe i etniczne:")
print(df["race_ethnicity"].unique())

# Liczba unikatowych grup
print("Liczba grup:", df["race_ethnicity"].nunique())

# Liczba rekordów dla każdej grupy
display(df["race_ethnicity"].value_counts())
```

```
Liczba rekordów: 598500
Lata w zbiorze:
[np.int64(2002), np.int64(2003), np.int64(2004), np.int64(2005), np.int64(2006), np.int64(2007), np.int64(2008), np.int64(20
Grupy rasowe i etniczne:
['American Indian, Alaska Native (non-Hispanic)'
 'Asian, Native Hawaiian, Pacific Islander (non-Hispanic)'
 'Black (non-Hispanic)' 'Hispanic' 'Multiple races (non-Hispanic)' 'Total'
 'White (non-Hispanic)']
Liczba grup: 7
```

race_ethnicity	count
American Indian, Alaska Native (non-Hispanic)	85500
Asian, Native Hawaiian, Pacific Islander (non-Hispanic)	85500
Black (non-Hispanic)	85500
Hispanic	85500
Multiple races (non-Hispanic)	85500
Total	85500
White (non-Hispanic)	85500

dtype: int64

```
# 10 rekordów z najniższymi wartościami val
df_sorted_asc = df.sort_values(by="val", ascending=True)
display(df_sorted_asc.head(10))

# 10 rekordów z najwyższymi wartościami val
df_sorted_desc = df.sort_values(by="val", ascending=False)
display(df_sorted_desc.head(10))
```

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care	spe
226566	2009	8	15 to 19	2	Female	Multiple races (non-Hispanic)	all	All conditions	Nursing Facility Care	S ca
56349	2004	11	30 to 34	3	Both	White (non-Hispanic)	resp_asthma	Asthma	Nursing Facility Care	S ca
56353	2004	11	30 to 34	3	Both	White (non-Hispanic)	resp_copd	COPD	Emergency Department Care	S ca
56355	2004	11	30 to 34	3	Both	White (non-Hispanic)	resp_copd	COPD	Nursing Facility Care	S ca
56356	2004	11	30 to 34	3	Both	White (non-Hispanic)	resp_copd	COPD	Prescribed Pharmaceuticals	S ca
56361	2004	11	30 to 34	3	Both	White (non-Hispanic)	rf_hypertension	Hypertension	Nursing Facility Care	S ca
424700	2005	8	15 to 19	1	Male	Multiple races (non-Hispanic)	cvd	Cardiovascular diseases	Nursing Facility Care	S ca
424701	2005	8	15 to 19	1	Male	Multiple races (non-Hispanic)	cvd	Cardiovascular diseases	Prescribed Pharmaceuticals	S ca
424704	2005	8	15 to 19	1	Male	Multiple races (non-Hispanic)	cvd_stroke	Cerebrovascular disease	Emergency Department Care	S ca
414564	2006	7	10 to 14	1	Male	American Indian, Alaska Native (non-Hispanic)	cvd_stroke	Cerebrovascular disease	Prescribed Pharmaceuticals	S ca

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care	spending_unit	
194263	2016	22	All Ages	3	Both	Total	all	All conditions	Total	Spending in 2016 USD (millions)	2440
193920	2015	22	All Ages	3	Both	Total	all	All conditions	Total	Spending in 2016 USD (millions)	2354
193577	2014	22	All Ages	3	Both	Total	all	All conditions	Total	Spending in 2016 USD (millions)	2244
193234	2013	22	All Ages	3	Both	Total	all	All conditions	Total	Spending in 2016 USD (millions)	2172
192891	2012	22	All Ages	3	Both	Total	all	All conditions	Total	Spending in 2016 USD (millions)	2145
192548	2011	22	All Ages	3	Both	Total	all	All conditions	Total	Spending in 2016 USD (millions)	2100
192205	2010	22	All Ages	3	Both	Total	all	All conditions	Total	Spending in 2016 USD (millions)	2069
191862	2009	22	All Ages	3	Both	Total	all	All conditions	Total	Spending in 2016 USD (millions)	2019
191519	2008	22	All Ages	3	Both	Total	all	All conditions	Total	Spending in 2016 USD (millions)	1936
191176	2007	22	All Ages	3	Both	Total	all	All conditions	Total	Spending in 2016 USD (millions)	1885

```
# 10 największych wartości w 2016 roku
df_top10_2016 = (
    df[df["year"] == 2016]
    .nlargest(10, "val")
)
```

```
display(df_top10_2016)

# 10 najmniejszych wartości w 2016 roku
df_bottom10_2016 = (
    df[df["year"] == 2016]
    .nsmallest(10, "val")
)

display(df_bottom10_2016)

# 10 największych wartości dla grupy Total w 2016 roku
df_top10_total_2016 = (
    df[
        (df["year"] == 2016) &
        (df["race_ethnicity"] == "Total")
    ]
    .nlargest(10, "val")
)

display(df_top10_total_2016)
```

Kolejne kroki: Wygeneruj kod za pomocą zmiennej `df_top10_2016` New interactive sheet Wygeneruj kod za pomocą zmiennej `df_bottom10_2016`

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care	spending_unit	
194263	2016	22	All Ages	3	Both	Total	all	All conditions	Total	Spending in 2016 USD (millions)	2
194312	2016	22	All Ages	3	Both	White (non-Hispanic)	all	All conditions	Total	Spending in 2016 USD (millions)	1
393763	2016	22	All Ages	2	Female	Total	all	All conditions	Total	Spending in 2016 USD (millions)	1
593263	2016	22	All Ages	1	Male	Total	all	All conditions	Total	Spending in 2016 USD (millions)	1
393812	2016	22	All Ages	2	Female	White (non-Hispanic)	all	All conditions	Total	Spending in 2016 USD (millions)	1
194257	2016	22	All Ages	3	Both	Total	all	All conditions	Ambulatory Care	Spending in 2016 USD (millions)	
194260	2016	22	All Ages	3	Both	Total	all	All conditions	Inpatient Care	Spending in 2016 USD (millions)	
593312	2016	22	All Ages	1	Male	White (non-Hispanic)	all	All conditions	Total	Spending in 2016 USD	

```
# Średnie wydatki według roku
mean_by_year = df.groupby("year")["val"].mean()

display(mean_by_year)
```

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care	spending_unit	
393757	2016	22	All Ages	2	Female	Total	all	All conditions	Ambulatory Care	Spending in 2016 USD (millions)	

year

	val
2002	1005.649585
2003	1054.361474
2004	1093.685502
2005	1123.613081
2006	1151.246796
2007	1178.491657
2008	1205.023960
2009	1253.341468
2010	1280.253397
2011	1293.395028
2012	1314.847846
2013	1325.203320
2014	1364.366702
2015	1429.304687
2016	1481.290078

dtype: float64

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care	spending_unit
4254	2016	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	rf_hypertension	Hypertension	Nursing Facility Care	Spen 201 (n
4402	2016	5	1 to 4	3	Both	Multiple races (non-Hispanic)	cvd_stroke	Cerebrovascular disease	Nursing Facility Care	Spen 201 (n
4408	2016	5	1 to 4	3	Both	Multiple races (non-Hispanic)	diabetes	Diabetes mellitus	Nursing Facility Care	Spen 201 (n
8746	2016	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	cvd_stroke	Cerebrovascular disease	Prescribed Pharmaceuticals	Spend capita
8752	2016	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	resp_asthma	Asthma	Nursing Facility Care	Spend capita
8758	2016	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	resp_copd	COPD	Inpatient Care	Spend capita
8764	2016	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	resp_copd	COPD	Prescribed Pharmaceuticals	Spend capita
8767	2016	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	rf_hypertension	Hypertension	Emergency Department	Spend capita

```
mean_by_race = df.groupby("race_ethnicity")["val"].mean()

display(mean_by_race)
```

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care	spending_unit
8769	2016	5	1 to 4	3	Both	American Indian, Alaska Native (non-Hispanic)	rf_hypertension	Hypertension	Nursing Facility Care	Spend capita

	year	age_group_id	age_group_name	sex_id	sex	race_ethnicity	acause	acause_name	type_care	spending_uni
194263	2016	22	All Ages	3	Both	Total	all	All conditions	Total	Spending 2016 US (millions
393763	2016	22	All Ages	2	Female	Total	all	All conditions	Total	Spending 2016 US (millions

593263	2016	22	All Ages	1	Male	Total	all	All conditions	Total	Spending 2016 US (millions

val

race_ethnicity	
American Indian, Alaska Native (non-Hispanic)	265.172879
Asian, Native Hawaiian, Pacific Islander (non-Hispanic)	253.297060
Black (non-Hispanic)	689.078641
Hispanic	524.863200
Multiple races (non-Hispanic)	316.768491
Total	3755.509251
White (non-Hispanic)	2853.878616

dtype: float64

						Total	all	All conditions	Ambulatory Care	Spending 2016 US (millions
						Total	all	All conditions	Inpatient Care	Spending 2016 US (millions
						Total	all	All conditions	Ambulatory Care	Spending 2016 US (millions
						Total	all	All conditions	Inpatient Care	Spending 2016 US (millions

```
agg_by_race = df.groupby("race_ethnicity")["val"].agg(
    ["mean", "count", "median"]
)

display(agg_by_race)
```

194262	2016	22	All Ages	3	Both	Total	all	All conditions	Prescribed Pharmaceuticals	Spending 2016 US (millions

race_ethnicity	mean	count	median
American Indian, Alaska Native (non-Hispanic)	265.172879	85500	7.000
Asian, Native Hawaiian, Pacific Islander (non-Hispanic)	253.297060	85500	10.635
Black (non-Hispanic)	689.078641	85500	46.140
Hispanic	524.863200	85500	31.180
Multiple races (non-Hispanic)	316.768491	85500	12.435
Total	3755.509251	85500	93.000
White (non-Hispanic)	2853.878616	85500	62.370

Kolejne kroki: Wygeneruj kod za pomocą zmiennej agg_by_race | New interactive sheet

```
multi_group = df.groupby(
    ["year", "race_ethnicity"]
)["val"].mean()

display(multi_group)
```

		val
2002	**American Indian, Alaska Native (non-Hispanic)**	247.272467
	Asian, Native Hawaiian, Pacific Islander (non-Hispanic)	205.501889
	Black (non-Hispanic)	559.617925
	Hispanic	393.325851
	Multiple races (non-Hispanic)	291.439346
...	**...**	...
2016	**Black (non-Hispanic)**	782.043353
	Hispanic	692.492937
	Multiple races (non-Hispanic)	393.035814
	Total	4559.435056
	White (non-Hispanic)	3350.492663

105 rows × 1 columns

dtype: float64

```
pivot = pd.pivot_table(
    df,
    values="val",
    index="year",
    columns="race_ethnicity",
```

```
    aggfunc="mean"
)

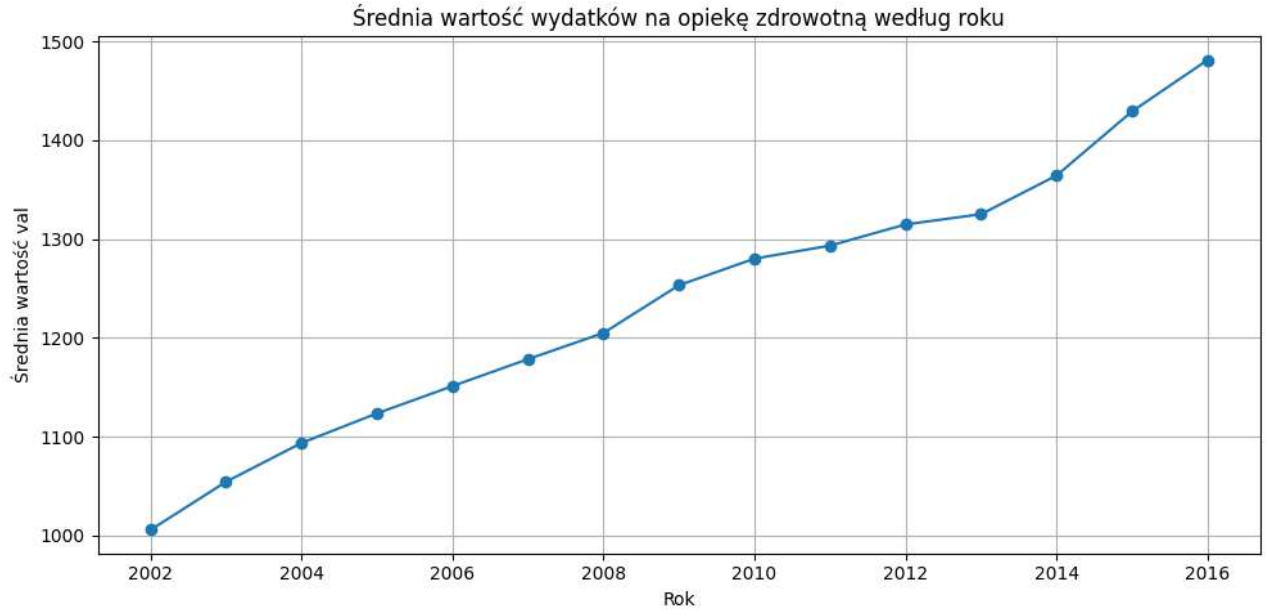
display(pivot)
```

race_ethnicity year	American Indian, Alaska Native (non-Hispanic)	Asian, Native Hawaiian, Pacific Islander (non-Hispanic)	Black (non-Hispanic)	Hispanic	Multiple races (non-Hispanic)	Total	White (non-Hispanic)
2002	247.272467	205.501889	559.617925	393.325851	291.439346	2980.122839	2362.266777
2003	242.474711	213.038800	590.253339	413.239796	290.468863	3146.805430	2484.249382
2004	246.054989	220.521381	614.225951	429.978023	291.920926	3277.576826	2575.520419
2005	249.515330	227.231274	633.467300	444.655918	292.974907	3378.120200	2639.326642
2006	251.441935	233.458472	650.015744	459.120065	294.387689	3473.323409	2696.980256
2007	254.135628	239.637614	666.382984	475.745881	295.918000	3566.473302	2751.148188
2008	259.202837	246.434346	681.025074	491.912700	299.072374	3654.607126	2802.913265
2009	271.759633	256.988125	709.561109	520.143095	309.457995	3806.357358	2899.122960
2010	278.394807	262.835412	724.861811	540.826523	314.455002	3894.544267	2945.855960
2011	279.733265	265.530153	729.628360	556.451670	317.707333	3942.275132	2962.439284
2012	281.023298	270.933153	738.011488	578.417918	324.772891	4015.044693	2995.731482
2013	275.861307	273.856482	737.643135	594.086023	328.372011	4056.467498	3010.136786
2014	275.582809	281.832596	748.770328	621.798496	341.612375	4186.587919	3094.382388
2015	279.871875	295.393761	770.671718	660.753107	365.931839	4394.897716	3237.612791
2016	285.268288	306.262437	782.043353	692.492937	393.035814	4559.435056	3350.492663

Kolejne kroki: Wygeneruj kod za pomocą zmiennej `pivot` New interactive sheet

```
# Wykres średniej wartości wydatków według roku
plt.figure(figsize=(10, 5))
mean_by_year.plot(kind="line", marker="o")

plt.title("Średnia wartość wydatków na opiekę zdrowotną według roku")
plt.xlabel("Rok")
plt.ylabel("Średnia wartość val")
plt.grid(True)
plt.tight_layout()
plt.show()
```




```
pivot.plot(figsize=(12, 6), marker="o")

plt.title("Średnia wartość wydatków według roku i grupy rasowej lub etnicznej")
plt.xlabel("Rok")
plt.ylabel("Średnia wartość val")
```

```
plt.legend(title="Grupa", bbox_to_anchor=(1.05, 1), loc="upper left")
plt.grid(True)
plt.tight_layout()
plt.show()
```

